

Human Immunodeficiency Virus 1+2 (HIV) Antibody (HIV(1+2)-Ab) ELISA Kit

Catalog No.: abx364861

Size: 96T

Range: Qualitative

Sensitivity: Qualitative

Storage: Store at 4°C for up to 6 months.

Application: For quantitative detection of HIV(1+2)-Ab in Serum and plasma.

Introduction: The human immunodeficiency virus (HIV) is a lentivirus (a subgroup of retrovirus) that causes HIV infection and over time acquired immunodeficiency syndrome (AIDS). AIDS is a condition in humans in which progressive failure of the immune system allows life-threatening opportunistic infections and cancers to thrive. Without treatment, average survival time after infection with HIV is estimated to be 9 to 11 years, depending on the HIV subtype. In most cases, HIV is a sexually transmitted infection and occurs by contact with or transfer of blood, pre-ejaculate, semen, and vaginal fluids. Non-sexual transmission can occur from an infected mother to her infant through breast milk. An HIV-positive mother can transmit HIV to her baby both during pregnancy and childbirth due to exposure to her blood or vaginal fluid. Within these bodily fluids, HIV is present as both free virus particles and virus within infected immune cells.

Principle of the Assay

This kit is based on indirect enzyme-linked immuno-sorbent assay technology. The 96-well plate is pre-coated with the target antigen. Samples are added to the wells and incubated, then washed with wash buffer. Samples that contain HIV(1+2)-Ab will bind to the pre-coated antigen to form an antigen-antibody complex. Free components are washed away with wash buffer. Next, HRP-conjugated antigen is added to the wells and incubated, then washed. TMB substrate is used to visualize HRP activity. TMB is catalyzed by HRP to produce a blue colour product that changes into yellow after adding stop solution. The intensity of the color yellow is proportional to the amount of HIV(1+2)-Ab bound on the plate. The O.D. absorbance is measured spectrophotometrically at 450 nm in a microplate reader, and then the concentration of HIV(1+2)-Ab can be calculated.

Kit components

1. One pre-coated 96-well microplate (12 × 8 well strips)
2. HIV-1 Positive control: 1 ml
3. HIV-2 Positive control: 1 ml
4. Negative control: 1 ml
5. HRP-conjugated antibody: 6 ml
6. Concentrated Wash buffer (20X): 50 ml
7. TMB substrate A: 6 ml
8. TMB substrate B: 6 ml
9. Stop solution: 6 ml
10. Plate Sealer: 3
11. Hermetic bag: 1

Material Required But Not Provided

1. 37°C incubator
2. Microplate reader (wavelength: 450 nm)
3. Multi and single channel pipettes and sterile pipette tips
4. Squirt bottle or automated microplate washer
5. ELISA shaker
6. Distilled or deionized water
7. 1.5 ml tubes to prepare standard/sample dilutions
8. Absorbent filter papers
9. 100 ml and 1 liter graduated cylinders

Protocol

A. Preparation of sample and reagents

1. Sample

Store samples to be assayed within 24 hours at 2-8°C. Alternatively, aliquot and store at -20°C or -80°C for long term. Avoid repeated freeze-thaw cycles.

- **Serum:** Samples should be collected into a serum separator tube. Coagulate the serum by leaving the tube undisturbed in a vertical position overnight at 4°C or at room temperature for up to 60 minutes. Centrifuge at approximately 1000 × g for 15 min. Analyze the serum immediately or aliquot and store at -20°C or -80°C.
- **Plasma:** Collect plasma using EDTA as an anticoagulant. Centrifuge for 15 minutes at 1000 x g within 30 minutes of collection. Assay immediately or aliquot and store at -20°C. Avoid hemolysis and high cholesterol samples.

Note:

- » Fresh samples or recently obtained samples are recommended to prevent degradation and denaturalization that may lead to erroneous results. It is recommended to store samples to be used within 5 days at 4°C, within 1 month at -20°C and within 2 months at -80°C.
- » Samples should be clear and transparent. Samples must be diluted so that the expected concentration falls within the kit's range.
- » Please bring sample slowly to room temperature. Sample hemolysis will influence the result. Hemolyzed specimen should not be used. Samples that contain NaN₃ cannot be detected as it interferes with HRP.
- » Always use non-pyrogenic, endotoxin-free tubes for blood collection.

2. Wash buffer

Dilute the concentrated Wash buffer 20-fold (1/20) with distilled water (i.e. add 25 ml of concentrated wash buffer into 475 ml of distilled water).

B. Assay Procedure

Equilibrate the kit components and samples to room temperature for at least 30 minutes prior to use.

1. Determine the number of wells to be used. Any strips that are not being used should be kept dry and stored at 4°C.
2. Set up three Negative Control wells with 50 µl of Negative Control per well. Set up two Positive Control wells, one with 50 µl of HIV-1 Positive Control and the other with 50 µl of HIV-2 Positive Control. Set up one blank well with no solution. Add 50 µl of sample to each sample well. Add the solution at the bottom without touching the sides of the well. Shake the plate gently to mix the contents.
3. Seal the plate with a cover and incubate at 37°C for 60 min.
4. Remove the cover and discard the plate contents by tapping the plate on absorbent filter papers or other absorbent material.
5. Wash the plate 5 times with wash buffer. Do not let the wells completely dry at any time.

Manual Washing: Discard the solution without touching the side walls and wash the plate five times. Fill each well completely with Wash buffer and incubate on an ELISA shaker for 2 min. Invert the plate each time and decant the contents; tap it 4-5 times on absorbent paper towel to completely remove the liquid.

Automated Washing: Discard the solution and wash the plate five times overfilling the wells with Wash buffer. After the final wash invert the plate, decant the contents; tap it 4-5 times on absorbent paper towel to completely remove the liquid. It is recommended that the washer be set for a soaking time of 1-2 min.

6. Add 50 µl of HRP-conjugate working solution into each well (except the blank wells). Add the solution at the bottom of each well without touching the side wall.
7. Seal the plate with a cover and incubate at 37°C for 30 min.
8. Wash the plate 5 times with wash buffer. Do not let the wells completely dry at any time.

9. Add 50 µl of TMB substrate A and 50 µl of TMB substrate B into each well. Cover the plate and incubate at 37°C in dark conditions for 15-20 minutes (incubation time is for reference only, do not exceed 30 minutes). When an apparent gradient appears in the positive control wells the reaction can be terminated.
10. Add 50 µl of Stop solution into each well (including the blank well). There should be a color change to yellow. Gently tap the plate to ensure thorough mixing.
10. Ensure that there are no fingerprints or water on the bottom of the plate, and that the fluid in the wells is free of bubbles. Measure the absorbance at 450 nm immediately.

C. Analysis

1. Calculations:

Mean absorbance of the positive control should be ≥ 0.80 .

Mean absorbance of the negative control should be ≤ 0.08 .

CUT OFF value (Negative control < 0.05) = $0.05 + 0.10 = 0.15$

CUT OFF value (Negative control ≥ 0.05) = Negative control + 0.10

2. Interpretation of results:

If the positive control value is ≥ 0.80 , and negative control value is ≤ 0.08 , the test is valid, otherwise, the test is invalid.

Samples:

If O.D. of samples $<$ CUT OFF, the test samples are considered negative.

If O.D. of samples $\geq$ CUT OFF, the test samples are considered positive.

Precision:

Variable coefficient (CV%) $\leq 15\%$

D. Precautions

1. Before using the kit, centrifuge the tubes to bring down the contents trapped in the lid.
2. Avoid foaming or bubbles when mixing or reconstituting components.
3. Wash buffer may crystallize and separate. If this happens, please heat the tube to dissolve.
4. It is recommended measuring each controls and samples in duplicate or triplicate.
5. Do NOT let the plate dry out completely as this will inactivate the biological material on the plate.
6. Ensure plates are properly sealed or covered during incubation steps.
7. Complete removal of all solutions and buffers during wash steps is necessary for accurate measurement readings.
8. Do not reuse pipette tips and tubes to avoid cross contamination.
9. Do not use expired components or components from a different kit.
10. The TMB Substrate B solution is easily contaminated; work under sterile conditions when handling the TMB substrate B solution. TMB Substrate B solution should also be protected from light. Equilibrate the TMB substrate at room temperature prior to use. Unreacted substrate should be colorless or very light yellow in appearance. Aspirate the dosage needed with sterilized tips and do not dump the residual solution back into the vial.
11. A pilot experiment experiment with the controls and a small number of samples is recommended to determine if sample dilution is required.
12. Keep the plate dry after opening and before use.